

Task 9: The Stadium Panorama (Projective - Homography)

- **Scenario:** You have two photos from two different cameras covering a football field. They share a small overlapping area in the middle.
- **Task:** Select 4 matching points found in both images. Calculate the Homography matrix to mathematically project the second camera's view into the first camera's space, stitching them together into a seamless panorama.

Task 10: The Master Forger (Full Hierarchy Challenge)

- **Scenario:** You need to digitally insert a flat image of a painting into an empty, heavily angled picture frame hanging on a museum wall.
- **Task:** Traverse the entire hierarchy. Use a **Linear Scale** to shrink the painting, a **Rigid Transformation** to move it near the frame, and finally a **Projective Transformation** to warp the painting's 4 corners precisely into the 3D perspective of the angled frame.

Task 5: The Robot's Assembly Line (Rigid Transformation)

- **Scenario:** A robotic arm needs to pick up a microchip on a conveyor belt. The chip is rotated sideways and slightly off-center.
- **Task:** A Rigid Transformation preserves the exact shape of an object. Combine your Rotation and Translation matrices into a single 3×3 matrix. Apply it simultaneously to align the chip perfectly with the robot's gripper.

Task 6: The Architect's Blueprint (Similarity Transformation)

- **Scenario:** You imported a house blueprint into your software, but it is too small, rotated randomly, and placed in the wrong corner.
- **Task:** A Similarity Transformation preserves angles but changes size. Combine Scale, Rotation, and Translation into one single 3×3 matrix operation to fix the blueprint in one mathematical step.

Task 7: The Glitched Image (General Affine Transformation)

- **Scenario:** A corrupted data transmission resulted in a photograph being squished, slanted, rotated, and moved all at once.
- **Task:** You are given the starting and ending coordinates of three distinct landmarks in the photo. Solve a system of linear equations using these 3 points to find the 6 unknown values of a general Affine matrix, then fix the image.

Task 8: The Sidewalk Illusion (Projective - Perspective)

- **Scenario:** You took a photo of 3D chalk art on a sidewalk, but because you were standing in front of it, the far edge looks smaller than the close edge.
- **Task:** Pick 4 corners of the art that should mathematically form a perfect square. Build a perspective transformation matrix to warp the image, creating a top-down "birds-eye" view.

Lab 03 Tasks

Task 1: The Tiny Fingerprint (Linear - Scale)

- **Scenario:** You are a detective analyzing a digital fingerprint from a crime scene, but it is too small to read.
- **Task:** Write a script to manually build a 2x2 scaling matrix. Enlarge the fingerprint by 300% on the x-axis and 300% on the y-axis. Calculate how to keep the fingerprint perfectly centered on the screen after it grows.

Task 2: The Dizzy Satellite (Linear - Rotation)

- **Scenario:** A weather satellite bumped into space debris and took photos of a city rotated at a steep 45-degree angle.
- **Task:** Formulate a 2x2 rotation matrix using sine and cosine functions. Spin the image back to true north. Manually calculate the new image dimensions so the corners of the city do not get chopped off during the spin.

Task 3: The Fast Train Barcode (Linear - Shear)

- **Scenario:** A scanner took a photo of a barcode on a fast-moving train, causing the image to look slanted to the right.
- **Task:** Construct a 2x2 shear matrix to "push" the pixels horizontally. Shift the slanted lines back into a perfect rectangle so the barcode can be read correctly.

Task 4: The Hidden Treasure Map (Affine - Translation)

- **Scenario:** A digital map is misaligned. The "X marks the spot" is hidden off the screen, exactly 150 pixels to the left and 80 pixels up.
- **Task:** Upgrade your matrices from 2x2 to 3x3 to allow for movement. Create a translation matrix and apply it to shift every pixel, sliding the map safely back into the visible frame.